Practical No. 2

Aim: Implement DDL and DML queries with different clauses.

Theory:
Implementing DDL (Data Definition Language) and DML (Data Manipulation Language) queries with different clauses is a fundamental aspect of working with relational databases. This practical theory provides an overview of how to practically implement these queries with various clauses to create, modify, and manipulate database structures and data.

Queries:
1) List the customer numbers (customer_no) and names (name) of all customers.

```
mysql> SELECT CUSTOMER_NO, NAME FROM CUSTOMER;
+-------------+--------------+
| CUSTOMER_NO | NAME         |
+-------------+--------------+
|          10 | GARRY SMITH  |
|          20 | PATEL        |
|          30 | DRAKE        |
|          40 | BOB SMITH    |
|          50 | JAMES        |
|          60 | NORTON       |
|          70 | JOHN MICHAEL |
+-------------+--------------+
7 rows in set (0.00 sec)

mysql> #202203103510097
```

2) List all details of the product with a product number (product_no) of 121 and 136.

```
mysql> SELECT * FROM PRODUCT WHERE PRODUCT_NO = 121 AND PRODUCT_NO = 136;
Empty set (0.00 sec)

mysql> #202203103510097
```

3) List all details of depots with rep 5 as their rep(rep_no).

```
mysql> SELECT * FROM DEPOT WHERE REP_NO = 5;
+----------+----------+---------+--------+
| DEPOT_NO | LOCATION | ADDRESS | REP_NO |
+----------+----------+---------+--------+
|        5 | WALES    | UK      |      5 |
+----------+----------+---------+--------+
1 row in set (0.00 sec)

mysql> #202203103510097
```

4) List the product number (product_no) and description only of all products from supplier number 1005 (supplier_no).

```
mysql> SELECT PRODUCT_NO, DESCRIPTION FROM PRODUCT WHERE SUPPLIER_NO = 1005;
+------------+----------------+
| PRODUCT_NO | DESCRIPTION    |
+------------+----------------+
|        120 | REDUCER        |
|        124 | WIDGET REMOVER |
+------------+----------------+
2 rows in set (0.08 sec)

mysql> #200203103510097
```

5) List all details for all customers with names (name) starting from sm followed by 1 character followed by t followed by anything.)

```
mysql> SELECT *
    -> FROM CUSTOMER
    -> WHERE NAME LIKE 'SM_T%';
Empty set (0.00 sec)

mysql> #202203103510097
```

6) List all details for all orders with date_placed from 1-jan-2023 to 31-jan-12023).

```
mysql> SELECT *
    -> FROM CORDER
    -> WHERE DATE_PLACED BETWEEN '1993-01-01' AND '1993-01-31';
+-----------+-------------+-------------+----------------+
| CORDER_NO | CUSTOMER_NO | DATE_PLACED | DATE_DELIVERED |
+-----------+-------------+-------------+----------------+
|       200 |          20 | 1993-01-01  | 1993-01-04     |
|       201 |          40 | 1993-01-17  | 1993-01-20     |
|       202 |          20 | 1993-01-01  | 1993-01-04     |
|       205 |          70 | 1993-01-31  | 1993-02-03     |
+-----------+-------------+-------------+----------------+
4 rows in set (0.00 sec)

mysql> #202203103510097
```

7) List the sales rep number (rep_no), depot number and address for depots located at NORTH and address is UK.

```
mysql> SELECT REP_NO, DEPOT_NO, ADDRESS FROM DEPOT WHERE LOCATION = 'NORTH' and ADDRESS = 'UK';
+--------+----------+---------+
| REP_NO | DEPOT_NO | ADDRESS |
+--------+----------+---------+
|      1 |        1 | UK      |
+--------+----------+---------+
1 row in set (0.00 sec)

mysql> #202203103510097
```

8) Give the total number of items (quantity) in stock in all depots.

```
mysql> SELECT SUM(QUANTITY)
    -> FROM STOCK;
+---------------+
| SUM(QUANTITY) |
+---------------+
|           540 |
+---------------+
1 row in set (0.05 sec)

mysql> #202203103510097
```

9) Give the total number of items (order line quantity) which have been ordered with corder_no 200.

```
mysql> SELECT * FROM ONLINE WHERE CORDER_NO = 200;
+-----------+------------+----------+
| CORDER_NO | PRODUCT_NO | QUANTITY |
+-----------+------------+----------+
|       200 |        120 |        5 |
+-----------+------------+----------+
1 row in set (0.03 sec)

mysql> #202203103510097
```

10) List product descriptions in reverse alphabetical order.

```
mysql> SELECT DESCRIPTION
    -> FROM PRODUCT
    -> ORDER BY DESCRIPTION DESC;
+----------------+
| DESCRIPTION    |
+----------------+
| WIDGET REMOVER |
| SIZE WIDGET    |
| SIZE WIDGET    |
| REDUCER        |
| PLATE          |
| HANDLE         |
+----------------+
6 rows in set (0.00 sec)

mysql> #202203103510097
```

11) List the customer details with name ends with N.

```
mysql> SELECT *
    -> FROM CUSTOMER
    -> WHERE NAME LIKE '%N';
+-------------+--------+---------------+----------+--------------+
| CUSTOMER_NO | NAME   | ADDRESS       | DEPOT_NO | CREDIT_LIMIT |
+-------------+--------+---------------+----------+--------------+
|          60 | NORTON | SAN FRANSISCO |        5 |        17000 |
+-------------+--------+---------------+----------+--------------+
1 row in set (0.03 sec)

mysql> #202203103510097
```

12) List the customers details with a CustomerName that have "r" in the second position:

```
mysql> SELECT *
    -> FROM CUSTOMER
    -> WHERE SUBSTRING(NAME, 2, 1) = 'r';
+-------------+-------+---------+----------+--------------+
| CUSTOMER_NO | NAME  | ADDRESS | DEPOT_NO | CREDIT_LIMIT |
+-------------+-------+---------+----------+--------------+
|          30 | DRAKE | BRIXTON |        4 |         7000 |
+-------------+-------+---------+----------+--------------+
1 row in set (0.00 sec)

mysql> #202203103510097
```

13) List the customers with a CustomerName that starts with "N" and are at least 4 characters in length.

```
mysql> SELECT *
    -> FROM CUSTOMER
    -> WHERE NAME LIKE 'N%' AND LENGTH(NAME) >= 4;
+-------------+--------+---------------+----------+--------------+
| CUSTOMER_NO | NAME   | ADDRESS       | DEPOT_NO | CREDIT_LIMIT |
+-------------+--------+---------------+----------+--------------+
|          60 | NORTON | SAN FRANSISCO |        5 |        17000 |
+-------------+--------+---------------+----------+--------------+
1 row in set (0.00 sec)

mysql> #202203103510097
```

14) Find all suppliers with a City containing the pattern “ny”

```
mysql> SELECT *
    -> FROM SUPPLIER
    -> WHERE ADDRESS LIKE '%NY%';
+-------------+-----------+---------+
| SUPPLIER_NO | NAME      | ADDRESS |
+-------------+-----------+---------+
|        1002 | RINGWORLD | GERMANY |
|        1005 | SMITH     | GERMANY |
+-------------+-----------+---------+
2 rows in set (0.05 sec)

mysql> #202203103510097
```

15) Selects all customers with a City starting with “L”, followed by any character, followed by “n”, followed by any character, followed by “n”

```
mysql> SELECT *
    -> FROM CUSTOMER
    -> WHERE ADDRESS LIKE 'L_N_N%';
Empty set (0.00 sec)

mysql> #202203103510097
```

Conclusion: Implementing DDL and DML queries with different clauses is a fundamental skill for working with databases. This practical theory provides an overview of the key steps and considerations for effectively using these queries to create, modify, and manipulate database structures and data. Hands-on practice and continuous learning are essential for mastering these skills and becoming proficient in database management.